IP Addressing and Subnetting

Question 1: List IP address classes A, B, C, D and E, with one real-world example for each.

In the traditional classful IPv4 system, addresses are grouped into five classes according to their first octet. Class A, B and C were designed for different network sizes. Class D is used for multicast, and Class E is reserved for experimental or future use. Modern networks mostly use Classless Inter-Domain Routing (CIDR), but the class system is still useful for learning.

Class	First-octet range	Traditional default mask	Purpose	Real-world example
A	1–126*	255.0.0.0 (/8)	Very large networks	A large organization could historically receive a Class A network. Private example: 10.0.0.0/8 used for internal company networks.
B	128–191	255.255.0.0 (/16)	Medium-sized networks	A college could divide a private 172.16.0.0/16 address space among campus buildings.
C	192–223	255.255.255.0 (/24)	Small networks	A home router may use 192.168.1.0/24 for phones, laptops and smart TVs.
D	224–239	Not applicable	Multicast groups	A video or audio streaming system on a managed network may use a multicast group such as 239.1.1.1.
E	240–255**	Not applicable	Reserved/experimental	Not normally assigned to everyday user devices; it may appear in lab or research experiments rather than a normal home network.

** 0 is reserved for special use; 127 is reserved for loopback, so the commonly taught usable Class A first-octet range is 1–126.*
***The range 240–255 is class E in the traditional scheme, with some addresses having special reservations. Class D/E are not ordinary host-address classes.*

Question 2: For 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address and valid host range.

Given: IP address = 192.168.10.45; subnet mask = 255.255.255.0, which is /24 because the mask has 24 network bits and 8 host bits.

Step A — Find the network address

The first three octets are network bits and the last octet is the host part. Keep the network part and set the host part to 0:

```
IP: 192.168.10.45
Mask: 255.255.255.0
Network: 192.168.10.0
```

This is also the result of the bitwise AND operation between the IP address and subnet mask.

Step B — Find the broadcast address

Set all host bits to 1. With /24, the last octet becomes 255:

```
Broadcast address = 192.168.10.255
```

Step C — Find valid host addresses

The network address (.0) identifies the network, and the broadcast address (.255) sends traffic to all hosts on that subnet. These two addresses are not assigned to normal devices. Therefore, the valid host range is **192.168.10.1 to 192.168.10.254**.

Check the host count: 8 host bits gives 2^8 = 256 total addresses. Subtract 2 reserved addresses: 256 – 2 = **254 usable host addresses**.

Question 3: Create four separate subnets for a food-delivery startup using 10.0.0.0/24.

The startup needs separate networks for Delivery Partners, Restaurants, Customers and Admins. The original 10.0.0.0/24 network has 256 total addresses. To create four equal-sized subnets, borrow 2 bits from the host portion because $2^2 = 4$ subnets. The new prefix is /26 and the subnet mask is 255.255.255.192. Each subnet contains 64 addresses, with 62 usable host addresses.

Group	Network / CIDR	Subnet mask	Valid host range	Broadcast
Delivery Partners	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126	10.0.0.127
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190	10.0.0.191
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254	10.0.0.255

Calculation steps

- Required subnets = 4. Find bits to borrow: $2^n \geq 4$, so n = 2 bits.
- New prefix = original /24 + 2 = **/26**.
- Subnet mask: /26 = **255.255.255.192**.
- Host bits left = 32 – 26 = 6 bits.
- Addresses per subnet = $2^6 = 64$; usable hosts = 64 – 2 = **62**.
- Block size in the last octet = 256 – 192 = **64**, so network addresses begin at 0, 64, 128 and 192.

> *These are private IPv4 addresses, suitable for an internal network. In a real company, VLANs and firewall rules should also be used to keep admin systems separated from customer and restaurant devices; subnetting alone does not provide full security.*

Question 4: A college uses 172.16.0.0/16 and needs at least 20 subnets. Find the new mask and hosts per subnet.

Step A — Calculate the number of bits to borrow

Choose the smallest number n for which 2^n is at least 20:
$2^4 = 16$, which is not enough.
$2^5 = 32$, which is enough.
Therefore, borrow **5 bits** from the host portion.

Step B — Find the new prefix and subnet mask

Original prefix = /16. New prefix = 16 + 5 = **/21**.
A /21 mask has 8 bits in the first octet, 8 in the second, and 5 network bits in the third octet:
`11111111.11111111.11111000.00000000`
Decimal subnet mask = **255.255.248.0**.

Step C — Calculate hosts per subnet

IPv4 has 32 bits. Host bits = 32 – 21 = **11**.
Total addresses per subnet = $2^{11} = 2{,}048$.
Usable host addresses = 2,048 – 2 = **2,046 hosts per subnet**.

> ***Final answer:*** *The new subnet mask is **255.255.248.0 (/21)**. This provides **32 subnets** from the original /16 block, which is enough for the required 20, and each subnet has **2,046 usable host addresses**.*

Question 5: Explain why 192.168.1.300 is invalid and show what a valid address looks like.

The address **192.168.1.300** is invalid because IPv4 addresses contain four decimal octets, and each octet must be from 0 to 255. The last octet, 300, is greater than 255, so it cannot be represented as a single 8-bit octet.

If the device is meant to join a typical home subnet using the mask 255.255.255.0 (/24), a valid host address could be **192.168.1.25** or **192.168.1.100**, assuming that address is not already in use and is within the router's configured subnet. For the 192.168.1.0/24 subnet, the normal usable host range is **192.168.1.1 to 192.168.1.254**; 192.168.1.0 is the network address and 192.168.1.255 is the broadcast address.

Conclusion: Check that every IPv4 octet is between 0 and 255, and ensure the device's address, subnet mask and gateway match the network configuration. A syntactically valid address must also be unique on the local network.